Wasel Palestine
Progress Report

Advanced Software Engineering — RESTful APIs Project
Instructor: Dr. Amjad Abu-Hassan

Project Name	Wasel Palestine — Smart Mobility & Checkpoint Intelligence API
Technology Stack	Node.js, Express.js, PostgreSQL, Sequelize, JWT, Docker, Apidog, k6
Deployment	Docker + Docker Compose
Documentation Tools	GitHub Wiki, README, Apidog/API-Dog, k6 reports
Prepared By	Muath Al-Refa'I & Mohammad Al-Refa'i

1. Executive Summary

Wasel Palestine is a backend API platform designed to organize, verify, and expose mobility-related information such as checkpoints, road incidents, crowdsourced reports, route estimates, weather context, and regional alerts. The current project status is functional completion with final packaging in progress. Core backend features are implemented, manually tested, and prepared for final documentation and recorded demonstration.

2. Current Project Status

Area	Status	Notes
Backend API structure	**Completed**	Versioned REST endpoints implemented under /api/v1
Authentication	**Completed**	JWT login, refresh token flow, logout, current-user endpoint
Database schema	**Completed**	Relational PostgreSQL schema with Sequelize models
Checkpoint module	**Completed**	Create, list, and status update flow working
Incident module	**Completed**	Create, list, verify, and close endpoints prepared
Crowdsourced reports	**Completed**	Report submission, moderation, and voting supported
Audit logging	**Completed**	Moderation actions stored and retrievable
Route estimation	**Completed**	Routing + weather integration with fallback-safe handling
API documentation	**Completed**	API-Dog exports and link
Performance testing	**Completed**	k6 runnable scripts and results

3. Completed Work

- Created a versioned REST backend using Node.js and Express.js.
- Connected the backend to a PostgreSQL relational database through Sequelize.
- Implemented authentication using JWT access and refresh tokens.
- Added validation middleware and request validators for main endpoints.
- Implemented checkpoint, incident, report, subscription, and alert modules.
- Integrated external APIs for routing (OSRM) and weather (Open-Meteo).
- Implemented report moderation and audit logging for moderator/admin actions.
- Prepared Docker-based deployment for reproducible execution.
- Prepared Apidog documentation structure and k6 performance scripts.

4. Testing Progress

4.1 Manual API Testing

- Health endpoint tested successfully.
- User registration, login, auth/me, refresh, and logout flows tested.
- Checkpoint creation and listing tested with admin access.
- Incident creation and listing tested successfully.
- Report submission and moderation tested successfully.
- Route estimation tested successfully with route, weather, and incident metadata.
- Audit log retrieval tested using admin authorization.

4.2 Performance Testing

k6 scripts were prepared for read-heavy, write-heavy, mixed, spike, and soak testing scenarios. Initial write-heavy testing showed high failure rates caused by the abuse-prevention rate limiter, which was later made configurable for controlled performance testing. Mixed-load testing showed that the main backend remained stable, while route-estimation latency was influenced by third-party API calls.

5. Observed Issues and Mitigation

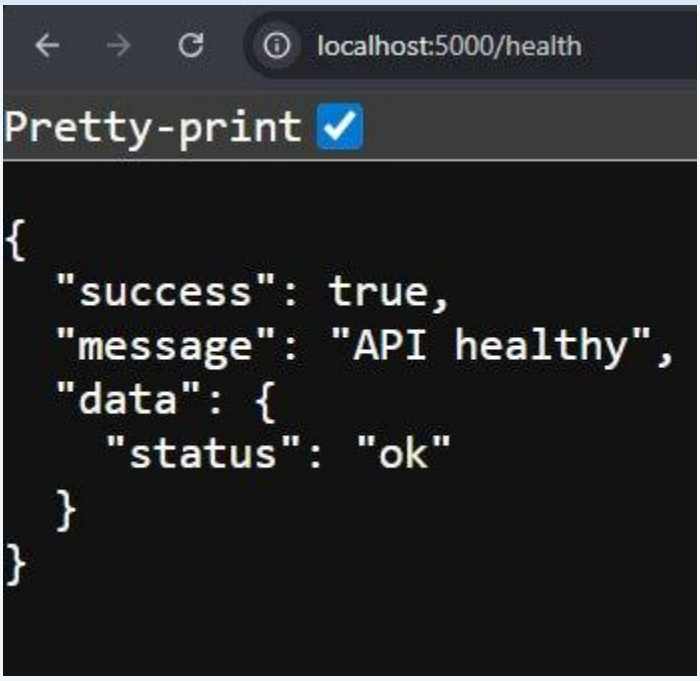
Issue	Observed Cause	Mitigation
High failure rate in write-heavy test	Report limiter blocked many requests	Rate limits were made configurable for controlled k6 testing
Slow p95 in mixed-load test	External route and weather APIs added latency	Route handling was made safer and can be tested separately at lighter load
Enum mismatch during seeding	Old seed values did not match shared enums	Seed data was updated to use CHECKPOINT_DELAY and ROAD_CLOSURE

6. Remaining Work Before Submission

- Export final Apidog documentation files (OpenAPI / Apidog export and environment file).
- Save final k6 output files and screenshots for the performance report.
- Finalize GitHub Wiki pages and make sure features are numbered consistently with Apidog/Postman.
- Push the final repository with feature branches and pull requests documented.
- Record the final demo and verify that the main endpoint flow runs without interruption.

7. Screenshot Evidence

System Health Check



Authentication/Login (Showing the bearer tokens)

Body Cookies Headers 2 Console Actual Request 200 97ms 679B Validation Success (200) Passed

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Login successful",
    "data": {
        "accessToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
        eyJpZCI6IjViMjNhMjAxLWM3ZTYtNDZkYi04ZmQ4LTFjOWVhYmY5NjQ2NiIsImVtYWlsIjoidGVzdEBleGFtcGxlLmNvbSIsInJvbGUiOiJVU0VSIiwiaWF0IjoxNzc3MTQ0MzcxLCJleHAiOjE3NzcxNDY
        xNzF9.VLbF857nkBYWqd7B7kIP5ubg80tgkmV_6nOF057Yj0w",
        "refreshToken": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
        eyJpZCI6IjViMjNhMjAxLWM3ZTYtNDZkYi04ZmQ4LTFjOWVhYmY5NjQ2NiIsImVtYWlsIjoidGVzdEBleGFtcGxlLmNvbSIsInJvbGUiOiJVU0VSIiwiaWF0IjoxNzc3MTQ0MzcxLCJleHAiOjE3NzcxNDk
        xNzF9.I5NM7_M7j50qTVYCEboyAfXV2JndzkvnTWzMURgWx6k",
        "tokenType": "Bearer"
    }
}
```


Checkpoint and Incident Management

Body Cookies Headers 2 Console Actual Request 200 25ms 436B

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Success",
    "data": {
        "items": [
            {
                "id": "c0342517-62b8-46eb-9ec1-c2e1f9b5c5cb",
                "name": "Beit Furik Checkpoint",
                "latitude": 32.176,
                "longitude": 35.335,
                "region": "Nablus",
                "statusCurrent": "OPEN",
                "description": "Normal movement",
                "createdAt": "2026-04-25T19:27:11.803Z",
                "updatedAt": "2026-04-25T19:27:11.803Z"
            }
        ],
        "total": 1,
        "page": 1,
        "size": 10
    }
}
```


Incident Management

Body Cookies Headers 2 Console Actual Request 201 36ms 640B

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Incident created",
    "data": {
        "id": "eb21e27b-5ade-4351-bb2e-bbb7f2b78574",
        "title": "Road closure near checkpoint",
        "description": "Temporary closure due to congestion",
        "category": "ROAD_CLOSURE",
        "severity": "HIGH",
        "status": "OPEN",
        "sourceType": "MANUAL",
        "locationLat": 32.123,
        "locationLng": 35.288,
        "region": "Nablus",
        "createdBy": "2a192f46-423c-472d-9e5b-251dd854d180",
        "updatedAt": "2026-04-25T19:29:03.508Z",
        "createdAt": "2026-04-25T19:29:03.508Z",
        "checkpointId": null,
        "verifiedBy": null,
        "verifiedAt": null,
        "closedBy": null,
        "closedAt": null
    }
}
```


Body Cookies Headers 2 Console Actual Request 200 25ms 1.65K

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Success",
    "data": {
        "items": [
            {
                "id": "eb21e27b-5ade-4351-bb2e-bbb7f2b78574",
                "title": "Road closure near checkpoint",
                "description": "Temporary closure due to congestion",
                "category": "ROAD_CLOSURE",
                "severity": "HIGH",
                "status": "OPEN",
                "sourceType": "MANUAL",
                "locationLat": 32.123,
                "locationLng": 35.288,
                "region": "Nablus",
                "checkpointId": null,
                "createdBy": "2a192f46-423c-472d-9e5b-251dd854d180",
                "verifiedBy": null,
                "verifiedAt": null,
                "closedBy": null,
                "closedAt": null,
                "createdAt": "2026-04-25T19:29:03.508Z",
                "updatedAt": "2026-04-25T19:29:03.508Z"
            },
            {
                "id": "070091a3-ccda-4d48-be75-500c6dc32fb4",
                "title": "Road closure near checkpoint",
                "description": "Temporary road closure near the checkpoint area",
                "category": "ROAD_CLOSURE",
```

Crowdsourced Report and Moderation

Body Cookies Headers 2 Console Actual Request 201 31ms 586B

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Report submitted",
    "data": {
        "report": {
            "id": "299aec91-ba7f-4814-8fa8-0d4f77ff7b2c",
            "abuseFlag": false,
            "confidenceScore": 0.5,
            "userId": null,
            "category": "CHECKPOINT_DELAY",
            "description": "Long waiting line reported by drivers",
            "latitude": 32.122,
            "longitude": 35.287,
            "region": "Nablus",
            "reportTime": "2026-04-17T16:30:00.000Z",
            "duplicateScore": 0.1,
            "moderationStatus": "PENDING",
            "updatedAt": "2026-04-25T19:35:04.696Z",
            "createdAt": "2026-04-25T19:35:04.696Z"
        },
        "possibleDuplicates": []
    }
}
```


Body Cookies Headers 2 Console Actual Request 200 36ms 571B

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Report moderated successfully",
    "data": {
        "report": {
            "id": "299aec91-ba7f-4814-8fa8-0d4f77ff7b2c",
            "userId": null,
            "category": "CHECKPOINT_DELAY",
            "description": "Long waiting line reported by drivers",
            "latitude": 32.122,
            "longitude": 35.287,
            "region": "Nablus",
            "reportTime": "2026-04-17T16:30:00.000Z",
            "moderationStatus": "APPROVED",
            "duplicateScore": 0.1,
            "abuseFlag": false,
            "confidenceScore": 0.5,
            "createdAt": "2026-04-25T19:35:04.696Z",
            "updatedAt": "2026-04-25T19:37:13.727Z"
        }
    }
}
```

Route Estimation

Body Cookies Headers 2 Console Actual Request 200 671ms 538B

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Route estimated",
    "data": {
        "estimatedDistanceKm": 18.28,
        "estimatedDurationMinutes": 51,
        "metadata": {
            "provider": "OSRM",
            "incidentCountAlongRoute": 3,
            "checkpointPenaltyApplied": true,
            "avoidAreas": [
                "Nablus"
            ],
            "notes": [
                "Checkpoint avoidance penalty applied",
                "Duration adjusted due to active incidents in selected region"
            ],
            "weather": {
                "time": "2026-04-25T19:30",
                "interval": 900,
                "temperature_2m": 18.1,
                "weather_code": 0,
                "wind_speed_10m": 4.6
            }
        }
    }
}
```

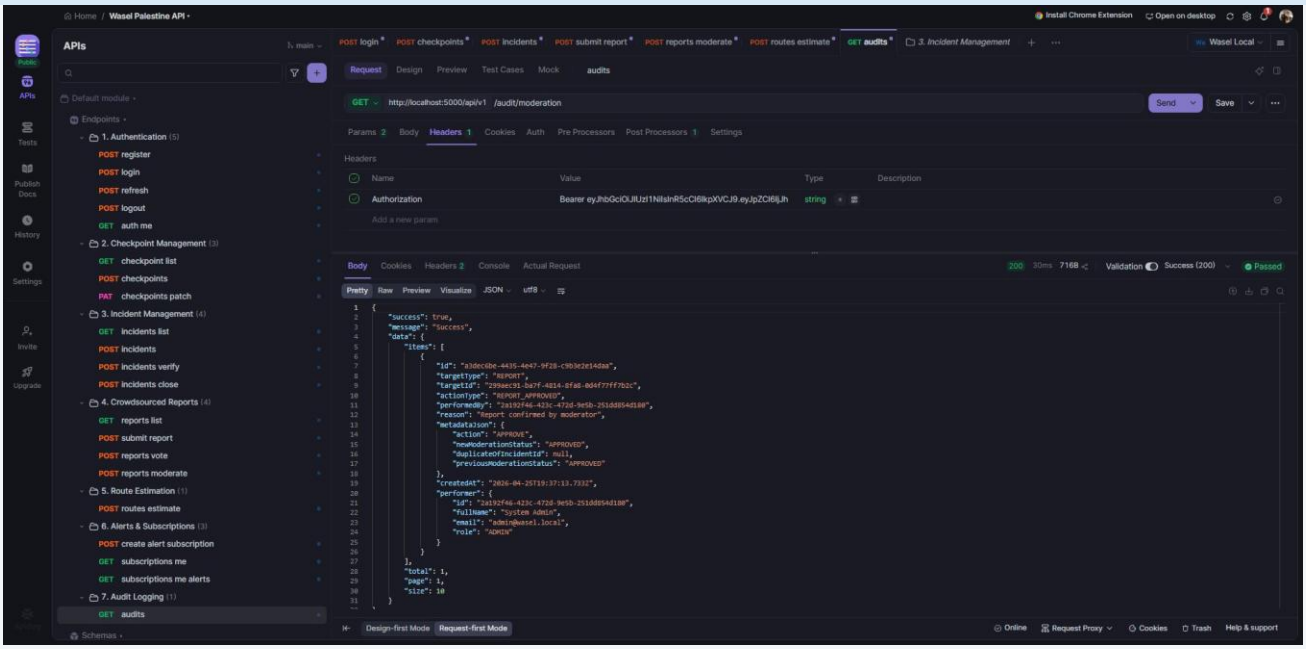

Audit Log Moderation

Body Cookies Headers 2 Console Actual Request 200 30ms 716B

Pretty Raw Preview Visualize JSON utf8


```
{
    "success": true,
    "message": "Success",
    "data": {
        "items": [
            {
                "id": "a3dec6be-4435-4e47-9f28-c9b3e2e14daa",
                "targetType": "REPORT",
                "targetId": "299aec91-ba7f-4814-8fa8-0d4f77ff7b2c",
                "actionType": "REPORT_APPROVED",
                "performedBy": "2a192f46-423c-472d-9e5b-251dd854d180",
                "reason": "Report confirmed by moderator",
                "metadataJson": {
                    "action": "APPROVE",
                    "newModerationStatus": "APPROVED",
                    "duplicateOfIncidentId": null,
                    "previousModerationStatus": "APPROVED"
                },
                "createdAt": "2026-04-25T19:37:13.733Z",
                "performer": {
                    "id": "2a192f46-423c-472d-9e5b-251dd854d180",
                    "fullName": "System Admin",
                    "email": "admin@wasel.local",
                    "role": "ADMIN"
                }
            }
        ],
        "total": 1,
        "page": 1,
        "size": 10
    }
```

API-Dog Documentation



k6 Performance Results

```
     execution: local
        script: performance/incident-list-read.js
        output: -

     scenarios: (100.00%) 1 scenario, 20 max VUs, 1m0s max duration (incl. graceful stop):
              * default: 20 looping VUs for 30s (gracefulStop: 30s)


  THRESHOLDS

    http_req_duration
    ✓ 'p(95)<500' p(95)=17.1ms

    http_req_failed
    ✓ 'rate<0.05' rate=0.00%


  TOTAL RESULTS

    checks_total.......: 1200    39.671671/s
    checks_succeeded...: 100.00% 1200 out of 1200
    checks_failed......: 0.00%   0 out of 1200

    ✓ status is 200
    ✓ has success true

    HTTP
    http_req_duration..............: avg=6.9ms min=544.4µs med=3.45ms max=88.19ms p(90)=9.03ms p(95)=17.1ms
      { expected_response:true }...: avg=6.9ms min=544.4µs med=3.45ms max=88.19ms p(90)=9.03ms p(95)=17.1ms
    http_req_failed................: 0.00%  0 out of 600
    http_reqs......................: 600    19.835836/s

    EXECUTION
    iteration_duration.............: avg=1s    min=1s      med=1s     max=1.09s   p(90)=1s     p(95)=1.01s
    iterations.....................: 600    19.835836/s
    vus............................: 20     min=20      max=20
    vus_max........................: 20     min=20      max=20

    NETWORK
    data_received..................: 1.2 MB 40 kB/s
    data_sent......................: 61 kB  2.0 kB/s



running (0m30.2s), 00/20 VUs, 600 complete and 0 interrupted iterations
default ✓ [======================================] 20 VUs  30s
```

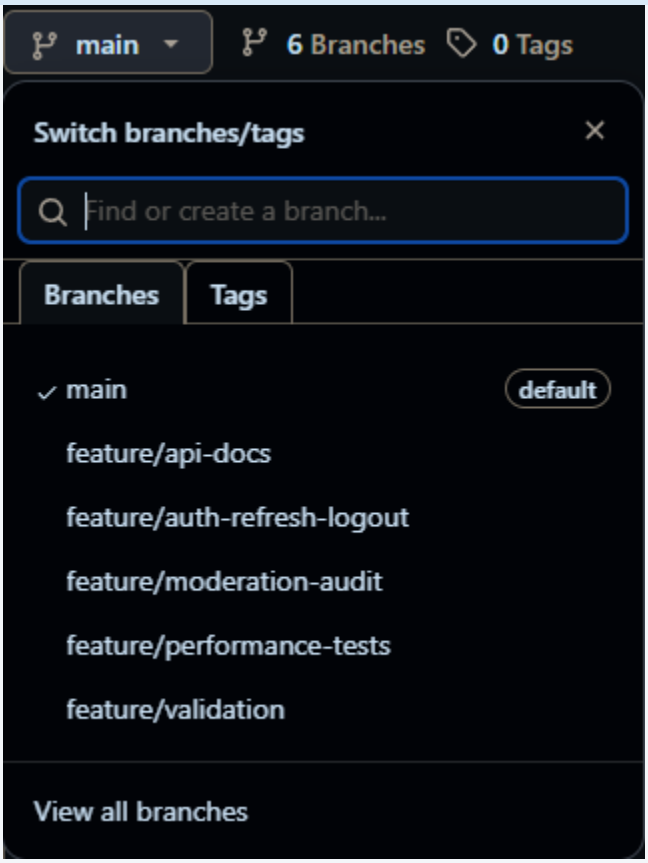
```
     execution: local
        script: performance/mixed-load.js
        output: -

     scenarios: (100.00%) 1 scenario, 20 max VUs, 1m15s max duration (incl. graceful stop):
              * default: 20 looping VUs for 45s (gracefulStop: 30s)



  THRESHOLDS

    http_req_duration
    ✓ 'p(95)<900' p(95)=13.54ms

    http_req_failed
    ✓ 'rate<0.10' rate=0.00%


  TOTAL RESULTS

    checks_total.......: 1800    39.72946/s
    checks_succeeded...: 100.00% 1800 out of 1800
    checks_failed......: 0.00%   0 out of 1800

    ✓ incident list 200
    ✓ incident list success
    ✓ report created
    ✓ report success

    HTTP
    http_req_duration..............: avg=5.68ms min=513.7µs med=3.95ms max=51.52ms p(90)=9.77ms p(95)=13.54ms
      { expected_response:true }...: avg=5.68ms min=513.7µs med=3.95ms max=51.52ms p(90)=9.77ms p(95)=13.54ms
    http_req_failed................: 0.00%  0 out of 900
    http_reqs......................: 900    19.86473/s

    EXECUTION
    iteration_duration.............: avg=1s     min=1s      med=1s     max=1.05s   p(90)=1.01s  p(95)=1.01s
    iterations.....................: 900    19.86473/s
    vus............................: 20     min=20      max=20
    vus_max........................: 20     min=20      max=20

    NETWORK
    data_received..................: 1.9 MB 42 kB/s
    data_sent......................: 148 kB 3.3 kB/s



running (0m45.3s), 00/20 VUs, 900 complete and 0 interrupted iterations
default ✓ [======================================] 20 VUs  45s
```

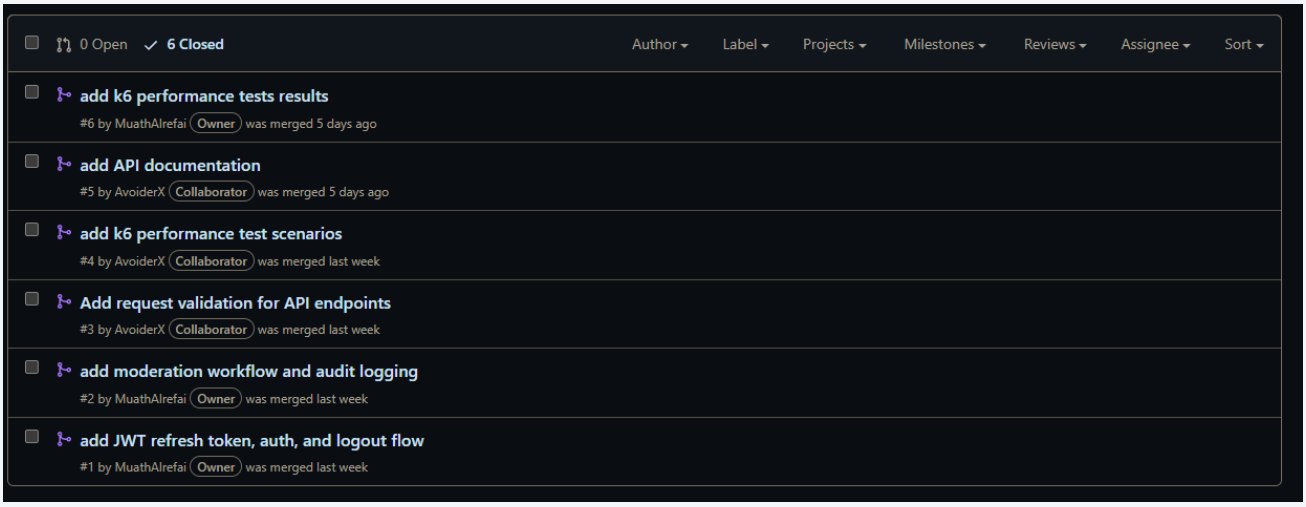
```
     execution: local
        script: performance/report-write.js
        output: -

     scenarios: (100.00%) 1 scenario, 10 max VUs, 1m0s max duration (incl. graceful stop):
              * default: 10 looping VUs for 30s (gracefulStop: 30s)


  THRESHOLDS

    http_req_duration
    ✓ 'p(95)<800' p(95)=37.15ms

    http_req_failed
    ✓ 'rate<0.10' rate=0.00%


  TOTAL RESULTS

    checks_total.......: 600     19.751201/s
    checks_succeeded...: 100.00% 600 out of 600
    checks_failed......: 0.00%   0 out of 600

    ✓ status is 201 or 200
    ✓ has success true

    HTTP
    http_req_duration..............: avg=11.08ms min=3.83ms med=6.66ms max=91.24ms p(90)=15.37ms p(95)=37.15ms
      { expected_response:true }...: avg=11.08ms min=3.83ms med=6.66ms max=91.24ms p(90)=15.37ms p(95)=37.15ms
    http_req_failed................: 0.00%  0 out of 300
    http_reqs......................: 300    9.8756/s

    EXECUTION
    iteration_duration.............: avg=1.01s   min=1s     med=1s     max=1.1s    p(90)=1.01s   p(95)=1.03s
    iterations.....................: 300    9.8756/s
    vus............................: 10     min=10       max=10
    vus_max........................: 10     min=10       max=10

    NETWORK
    data_received..................: 676 kB 22 kB/s
    data_sent......................: 94 kB  3.1 kB/s



running (0m30.4s), 00/10 VUs, 300 complete and 0 interrupted iterations
default ✓ [======================================] 10 VUs  30s
```

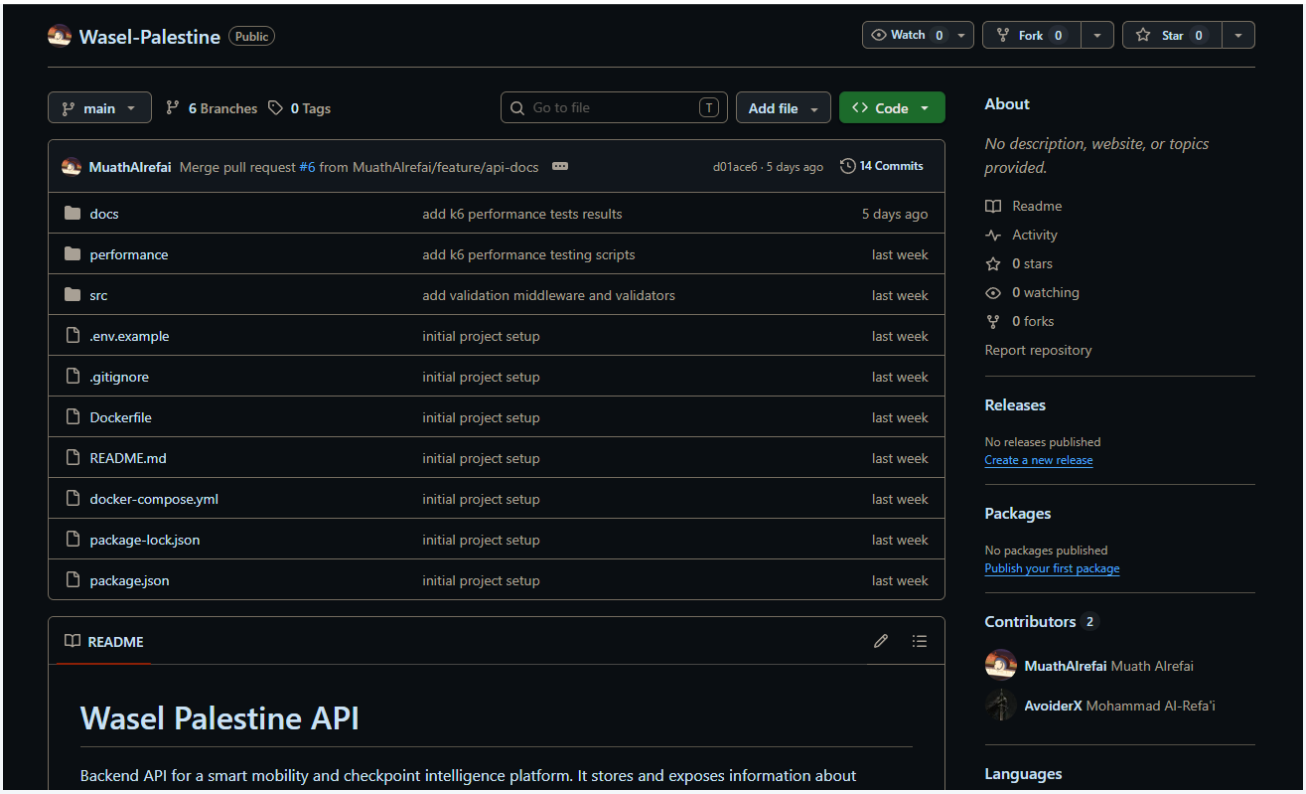
```
     execution: local
        script: performance/soak-test.js
        output: -

     scenarios: (100.00%) 1 scenario, 15 max VUs, 2m30s max duration (incl. graceful stop):
              * default: 15 looping VUs for 2m0s (gracefulStop: 30s)



  THRESHOLDS

    http_req_duration
    ✓ 'p(95)<700' p(95)=9.54ms

    http_req_failed
    ✓ 'rate<0.05' rate=0.00%


  TOTAL RESULTS

    checks_total.......: 1800    14.926221/s
    checks_succeeded...: 100.00% 1800 out of 1800
    checks_failed......: 0.00%   0 out of 1800

    ✓ status is 200

    HTTP
    http_req_duration..............: avg=4.16ms min=513.1µs med=3.15ms max=41.42ms p(90)=6.9ms p(95)=9.54ms
      { expected_response:true }...: avg=4.16ms min=513.1µs med=3.15ms max=41.42ms p(90)=6.9ms p(95)=9.54ms
    http_req_failed................: 0.00%  0 out of 1800
    http_reqs......................: 1800   14.926221/s

    EXECUTION
    iteration_duration.............: avg=1s     min=1s      med=1s     max=1.04s   p(90)=1s    p(95)=1.01s
    iterations.....................: 1800   14.926221/s
    vus............................: 15     min=15        max=15
    vus_max........................: 15     min=15        max=15

    NETWORK
    data_received..................: 3.6 MB 30 kB/s
    data_sent......................: 182 kB 1.5 kB/s



running (2m00.6s), 00/15 VUs, 1800 complete and 0 interrupted iterations
default ✓ [======================================] 15 VUs  2m0s
```

```
     execution: local
        script: performance/spike-test.js
        output: -

     scenarios: (100.00%) 1 scenario, 80 max VUs, 1m30s max duration (incl. graceful stop):
              * default: Up to 80 looping VUs for 1m0s over 5 stages (gracefulRampDown: 30s, gracefulStop: 30s)



  THRESHOLDS

    http_req_duration
    ✓ 'p(95)<1200' p(95)=6.37ms

    http_req_failed
    ✓ 'rate<0.15' rate=0.00%


  TOTAL RESULTS

    checks_total.......: 2638    43.822973/s
    checks_succeeded...: 100.00% 2638 out of 2638
    checks_failed......: 0.00%   0 out of 2638

    ✓ status is 200

    HTTP
    http_req_duration..............: avg=3.21ms min=504.9µs med=2.75ms max=25.46ms p(90)=4.99ms p(95)=6.37ms
      { expected_response:true }...: avg=3.21ms min=504.9µs med=2.75ms max=25.46ms p(90)=4.99ms p(95)=6.37ms
    http_req_failed................: 0.00%  0 out of 2638
    http_reqs......................: 2638   43.822973/s

    EXECUTION
    iteration_duration.............: avg=1s     min=1s      med=1s     max=1.02s   p(90)=1s     p(95)=1s
    iterations.....................: 2638   43.822973/s
    vus............................: 1      min=1         max=80
    vus_max........................: 80     min=80        max=80

    NETWORK
    data_received..................: 5.3 MB 88 kB/s
    data_sent......................: 266 kB 4.4 kB/s



running (1m00.2s), 00/80 VUs, 2638 complete and 0 interrupted iterations
default ✓ [======================================] 00/80 VUs  1m0s
```

GitHub Repository Workflow







8. Team Roles and Collaboration Notes

The project was completed collaboratively by two team members.

• Mauth Al-Refa'i: Focused mainly on the core backend implementation, including project structure, database models, authentication flow (register, login, refresh, logout), and the main checkpoint, incident, report, and subscription endpoints.

• Mohammad Al-Refa'i: Focused mainly on integration, testing, and delivery support, including external APIs (routing and weather), audit logging, API documentation in API-Dog, validation, k6 performance testing, Docker execution, wiki/report preparation, screenshots, and final demo packaging.

• Shared work: Both members participated in debugging, endpoint verification, review of responses, and final submission preparation.

Member Name	Student ID	Contribution
Muath Al-Refa'i	12011560	50%
Mohammad Al-Refa'i	12217414	50%

9. Conclusion

The project has reached a strong finalization stage. Core backend functionality is implemented and testable, the major documentation components are prepared, and the remaining effort is mainly focused on packaging, exports, screenshots, and recorded presentation.